

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 03

Configure AWS CloudTrail Management Event Logging

AWS CloudTrail | Management-Plane Telemetry | Amazon S3

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 8, 2026
STATUS	Complete

Executive Summary

Configured AWS CloudTrail to capture management events in the US West (Oregon) and US East (Ohio) regions and deliver logs to Amazon S3. The lab configuration intentionally did not use a customer-managed KMS key, reducing additional cost and friction while preserving a practical learning environment.

Business Problem

Without management-plane telemetry, administrators have limited ability to determine who changed a resource, which API action occurred, or when an account configuration was modified. The environment needed an audit trail before additional security and infrastructure services were added.

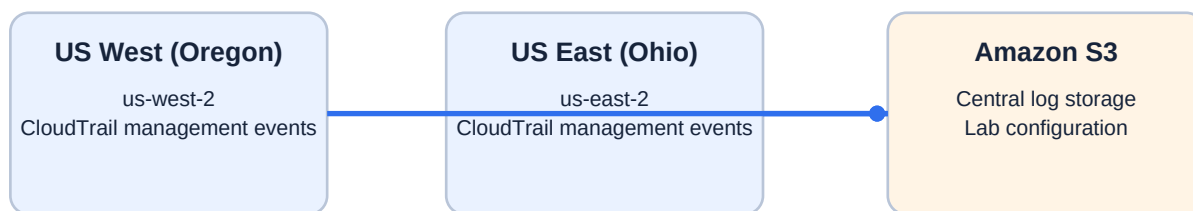
Objectives

- Capture AWS management events for administrative visibility.
- Configure logging for us-west-2 and us-east-2.
- Deliver CloudTrail log files to Amazon S3.
- Keep the learning environment cost-conscious.
- Establish a foundation for later monitoring and incident investigation.

Environment

Cloud Provider	Amazon Web Services (AWS)
Service	AWS CloudTrail
Regions	us-west-2 (Oregon) and us-east-2 (Ohio)
Event Type	Management events
Log Destination	Amazon S3
Customer-Managed KMS Key	Not enabled for this lab
Purpose	Audit logging and security visibility

Architecture



Management-plane activity is collected in two AWS regions and delivered to Amazon S3 for audit retention.

Figure 1 - CloudTrail management-event logging architecture.

Implementation Summary

1. Configured CloudTrail management-event logging for the US West (Oregon) region.
2. Configured CloudTrail management-event logging for the US East (Ohio) region.
3. Configured Amazon S3 as the log storage destination.
4. Did not enable a customer-managed KMS key to reduce lab cost and simplify log handling.
5. Established the logging baseline for future security monitoring and investigations.

EVIDENCE STATUS

No Project 03 console screenshot file was present among the uploaded evidence files. The document includes the complete implementation record and architecture; user-provided screenshots can be inserted in the next document revision without changing the standard format.

Security Analysis

CloudTrail provides management-plane telemetry that helps reconstruct administrative activity, investigate changes, and support security monitoring.

The lab did not enable a customer-managed KMS key. This is different from leaving logs unencrypted: Amazon S3 can still provide server-side encryption. The decision avoided additional KMS cost and operational friction in a learning environment.

Collecting management events in more than one region reduces visibility gaps when administration occurs outside a single primary region.

Lessons Learned

Cloud security investigations depend on management-plane visibility. Logging should be enabled before complex infrastructure is deployed so later changes can be reviewed.

Encryption decisions require precision. Disabling a customer-managed KMS key for a lab is not the same as intentionally storing plaintext logs.

Production Improvements

- Use an organization-wide multi-region trail when managing multiple accounts and regions.
- Use centralized log storage with restricted write and read permissions.
- Evaluate a customer-managed KMS key when compliance, separation of duties, or key control requirements justify it.
- Integrate CloudTrail with detection and alerting workflows.
- Define S3 lifecycle and retention policies for long-term cost control.

Skills Demonstrated

CloudTrail configuration	Management-event logging	Auditability
Multi-region administration	API activity visibility	Retention planning
Amazon S3 log storage	Incident investigation foundation	Cost-aware design

Resume Bullet

RESUME

Configured AWS CloudTrail management-event logging across us-west-2 and us-east-2 with Amazon S3 delivery, establishing an audit trail for administrative activity and future security monitoring.

STAR Interview Story

Situation: The AWS lab environment needed visibility into administrative actions and API activity.

Task: Create an audit logging foundation across the regions being used for lab work while keeping the design cost-conscious.

Action: I configured CloudTrail management-event logging in us-west-2 and us-east-2, delivered logs to Amazon S3, and did not enable a customer-managed KMS key to avoid extra lab cost and log-access friction.

Result: The environment gained an auditable record of management activity and a foundation for future monitoring, investigations, and governance work.

Version History

v1.0.0	July 8, 2026	Initial standardized portfolio release. Configured CloudTrail management-event logging for us-west-2 and us-east-2, delivered logs to Amazon S3, and did not enable a customer-managed KMS key in order to reduce lab cost and log-handling friction.

References

- Cloud Security Lab a Week (S.L.A.W.) - Turn on CloudTrail.
- Project implementation notes supplied by the project author.